

AIU SMART CAMPUS

Energy Impact & Benchmark Analysis

How much energy could an integrated cyber-physical control platform realistically save at Alamein International University, New Alamein — based on comparable university implementations and AIU's actual campus scale?

How to read this report

Every number is tagged by origin: **ACADEMIC EVIDENCE** (published results), **OFFICIAL AIU DATA** (contract / university documents), **ENGINEERING ESTIMATE** (our modelling), or **PROJECT TARGET**. Anything that could not be verified is marked **Not publicly verified**.

The AIU figures are a **benchmark-based estimated baseline** and **scenario projections** — not measured AIU consumption and not a promise of savings. They must be validated against AIU's own meter data (Section 13).

Graduation project — Cyber-Physical Smart Building Control System
Prepared October 2026

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Executive summary

Universities have already demonstrated measurable energy savings through occupancy sensing and automated control. The question for AIU is how much of that opportunity an integrated platform can capture, and how confidently that can be said before AIU's own meters are analysed.

Headline findings

Evidence exists. A peer-reviewed field study at H University (Seoul, South Korea) measured **77.6%** (parking lighting), **32.4%** (classroom lighting), **27.9%** (classroom cooling) and **28.2%** (dormitory) savings with motion sensors and a card-key room-management system. Programme-level and building-level savings reported by U.S. universities range from **9% to 61%**, but each number has a different scope.

Demonstrated range is wide, and scope matters. Savings of 9–20% come from campus/portfolio programmes and analytics; ~28–34% from end-use or building retrofits; 51–61% from laboratory ventilation redesign, which our system does not do.

AIU baseline (benchmark-based estimate, not measured): 263,020 m² of contractor-documented built-up area × 165 kWh/m²·yr ≈ **43.40 GWh/yr** (range 22.62–93.96 GWh/yr across area and EUI sensitivities).

Controllable energy ≠ total energy. Our system acts on lighting, HVAC set-points/shutdown, non-critical plug loads and scheduled devices. We model 50–70% of campus electricity as controllable, and 50–90% of that as covered by the system.

Scenario results (annual): Conservative **1.08 GWh** (2.5% of total); Moderate **3.28 GWh** (7.6%); High-impact **7.66 GWh** (17.6%).

Most defensible planning range for AIU: Conservative-to-Moderate — roughly **5–13% reduction in controllable energy**, equal to about **2.5–7.6% of total campus electricity** (1.08–3.28 GWh/yr; ≈ EGP 2.8–8.4 million/yr at EGP 2.55/kWh). The High-impact case is an upper bound that requires near-complete coverage and Korean-study-level results.

Honest limits. AIU's actual consumption, tariff category, current operating area and student numbers are not publicly verified. The 263,020 m² and 174,000 m² figures are contractor-stated BUA of construction packages — they are not the campus total and they are not the same thing (Section 5). The ≈150-acre campus area is stated in acres by the source we could access; it is a land area, not a built area.

1. Problem and approach

Campus buildings are occupied intermittently but are often conditioned and lit on fixed schedules. In hot climates, cooling dominates electricity use, so mismatches between occupancy and operation are costly. Published work shows that occupancy-driven control reduces this waste; what is rarely quantified is how that translates to a specific campus of a specific size.

This report therefore answers the question in three transparent steps, each reproducible from the model file delivered with the report:

1. **Extract measured results** from the Korean field study and verified university benchmarks, labelling what each number represents.
2. **Establish AIU's scale** from official and contractor documents, separating current, under-construction and planned areas.
3. **Apply a three-factor scenario model** — controllable share × automation coverage × saving rate on controlled loads — to a benchmark-based baseline, and present ranges instead of a single figure.

Chart 6 — AIU Smart Campus control pipeline

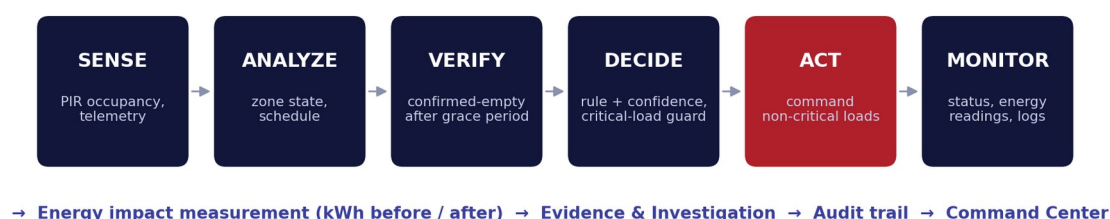


Chart 6 — AIU Smart Campus control pipeline (SENSE → ANALYZE → VERIFY → DECIDE → ACT → MONITOR)

2. The Korean university experimental study

Source and provenance

Lee, J.-W. & Kim, Y. I. (2020). *Energy Saving of a University Building Using a Motion Detection Sensor and Room Management System.* Sustainability 12(22), 9471. Seoul National University of Science and Technology. **This is a South Korean study at a Seoul campus ("H University"), not an American study.** The paper cites U.S. universities (e.g. Boston University, via Abidin et al.) only in its literature review, as institutions that had adopted occupancy sensors.

The authors measured electricity with and without sensors during normal operation rather than relying on simulation. Test objects: the Cyber Building (16,476 m², 27 classrooms and 35 labs, B2–8F, completed Aug 2013) and a female dormitory (5,994.53 m², 195 rooms, completed Mar 2019).

Experiment	Set-up and duration	Result (sensor OFF → ON)	Saving
1. Underground parking lot (B1 passageway)	918.66 m ² , 51 lights (2.22 W/m ²); RF/Doppler sensors dim to 20% and raise to 100% on detection; ~6 weeks, May–June; three on/off tests	50.9 → 11.4 Wh/m ² -day (saved 39.5)	77.6% (tests: 77.7 / 79.3 / 75.7)
2a. Lecture rooms – lighting	3 classrooms, 27 × 42 W LED; RF motion sensors switched on/off on alternate weeks, 4 weeks (26 May–22 Jun); outdoor temp 22.2–23.8 °C	77.14 → 52.12 Wh/m ² -day (saved 25.0)	32.4% (two tests: 37.9 / 27.8)

Experiment	Set-up and duration	Result (sensor OFF → ON)	Saving
2b. Lecture rooms – cooling (EHP)	3 EHP units, 11 kW rated cooling; same weeks	197.11 → 142.06 Wh/m ² ·day (saved 55.0)	27.9% (two tests: 33.3 / 23.5)
3. Dormitory (key-card room-management system, RMS)	10 test rooms vs 120 reference rooms (random from 195); 4 weeks baseline with RMS on, then 3 weeks with RMS off in test rooms; reference-group correction, Oct–Nov	Weekly per room: 68.83 kWh (RMS off) vs 49.45 kWh corrected (RMS on); saved 19.38 kWh/week (142.4 Wh/m ² ·day)	28.2% (weekly: 40.0 / 16.2 / 30.9)

What these numbers do and do not mean

- **Application-specific, not campus-wide.** 77.6% applies to parking-lot lighting only (an all-night, low-occupancy space with dimmable fixtures). It must not be extrapolated to classrooms or to total campus energy.
- **Short experiments.** Four to seven weeks per object, at one campus, in shoulder-season weather (May–June, Oct–Nov). Classroom results rest on two on/off pairs and the authors note that class operation, weather and occupancy could not be held identical week to week.
- **Dormitory saving is a calibrated estimate.** It is the avoided increase when the RMS was switched off (test rooms used 32.3% more than the reference group, with 10 rooms and large room-to-room variance). The RMS is key-card-based presence logic, not a PIR motion sensor.
- **Per-area figures are daily, in Wh/m²·day**, during the measured weeks; the paper does not provide annualised campus totals, and neither do we.
- The paper's own literature review adds context: occupancy sensor + smart plug ≈34% (Jeong & Seo); daylight + occupancy lighting up to 65.2% (Hong et al.); classroom lighting PIR 10% (Nam et al.); campus platform 15–30% (Kolokotsa et al.).

3. Comparable university implementations

Each row states what the saving actually measures. **Do not average these numbers:** they mix lighting-only, HVAC-only, whole-building, laboratory-specific and campus-programme results, and some are targets or estimates rather than measured outcomes.

University / project	Country	Scale	Technology	Application & scope	Saving	Energy / cost saved	Source
H University (Lee & Kim)	South Korea	Lecture bldg 16,476 m ² ; dorm 5,994.53 m ² ; parking 918.66 m ² test area	RF/Doppler motion sensors; key-card RMS	END-USE, measured before/after: lighting, EHP cooling, dormitory	77.6% (parking light); 32.4% (class light); 27.9% (class cooling); 28.2% (dorm)	39.5 / 25.0 / 55.0 / 142.4 Wh/m ² -day; cost not reported	S1
Michigan State Univ. — Anthony Hall	USA	1 building, 317,200 sq ft (≈29,470 m ²)	Airflow monitoring, economizer repair, VFDs, demand ventilation, heat recovery, lighting controls	BUILDING-LEVEL multi-measure retrofit (partner-reported)	34% (source EUI)	\$536,000/yr; kWh not stated	S2
UC Irvine — Smart Labs	USA	10 retrofitted labs; Natural Sciences II >100,000 sq ft	Variable ventilation + air-quality + occupancy sensors + digital controls	LABORATORY-specific; NS-II figure excludes lab process loads	61% avg (10 labs); 60% headline; 51% NS-II	NS-II \$180,000/yr	S3–S5
Pace University	USA	Portfolio 1–5 million sq ft	BMS scheduling, set-point policy, occupancy-matched HVAC, 2 energy specialists	CAMPUS-WIDE management programme (not a sensor product)	≈18% avg annual (secondary); GHG ~24%	26M kWh + ~1M therms; \$6.8M (thru Oct 2022)	S6, S7
Colorado Mesa University	USA	16 buildings, 1.2 million sq ft	District geothermal heat pumps (450 boreholes)	HVAC PLANT technology — not sensing/automation	≈ half of conventional HVAC energy (DOE)	14.86M kWh/yr; \$1.5M/yr; 10 kWh/sq ft-yr	S8
Univ. of Utah — Eyring buildings	USA	5 buildings, 370,000 sq ft	CAV→VAV + sensors/controls	LAB/CLASSROOM retrofit — ESTIMATED	44% EUI (est.)	\$456,000/yr (est.)	S9
Carnegie Mellon Univ. (DOE BENEFIT)	USA	Research buildings	Depth-sensing occupancy + model-predictive control	Research TARGET — not a result	20% target	—	S10
LBNL Smart Energy Analytics Campaign	USA (portfolio)	104 orgs; 6,500 bldgs; >0.5 bn sq ft	Fault detection & diagnostics; energy information systems	PORTFOLIO medians (office + higher-ed = 80%)	9% (FDD); 3% (EIS) median	—	S12
ACEEE smart-buildings review	USA (model)	Typical 100,000 sq ft education; 70,000 sq ft lab	Occupancy sensors + web lighting control; lab air-quality + occupancy + real-time ventilation	TECHNOLOGY-REVIEW model values	11% (education); 40% (lab)	20,900 and 390,000 kWh/yr per model building	S11
Technical Univ. of Crete (via S1)	Greece	Campus platform (CAMP-IT)	Web-based campus energy-management / control	CAMPUS platform (reported in S1's review)	15–30% (as cited)	—	S13

Scope taxonomy used in this report

Lighting-only / end-use: Korean parking and classroom lighting. **HVAC-only:** Korean classroom EHP cooling. **Building-level:** MSU Anthony Hall, Utah. **Laboratory-specific:** UC

Irvine, ACEEE lab. **Campus-/portfolio-level programme:** Pace, LBNL, Crete. **Different mechanism (excluded from rate selection):** Colorado Mesa geothermal. **Target / estimate (not measured):** CMU, Utah.

Observation: the more narrowly an end-use is controlled (lighting in a low-occupancy space), the higher the saving; the more broadly the metric is defined (whole campus or portfolio), the lower and more stable it becomes ($\approx 9\text{--}20\%$).

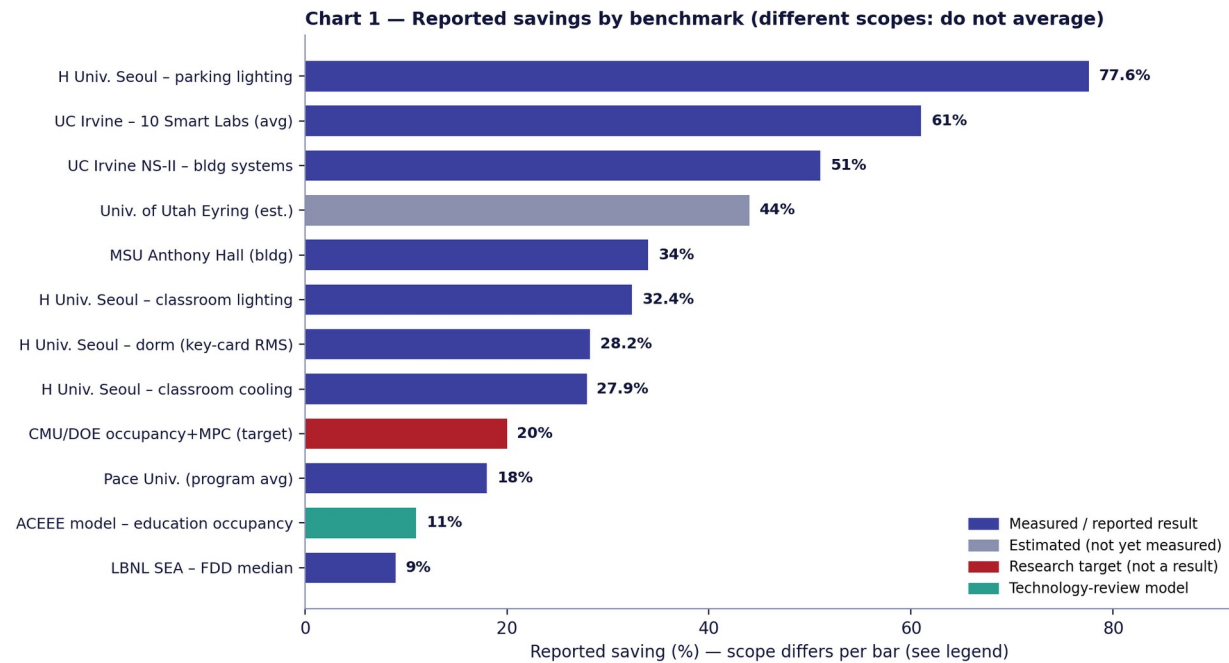


Chart 1 — Reported savings by benchmark (scopes differ; bars must not be averaged)

4. Classrooms versus laboratories

Laboratories behave very differently from classrooms, and Smart Lab 301 is the scenario our project demonstrates, so the two must be modelled separately.

	General classrooms / lecture rooms	Laboratories
Dominant load	Lighting and cooling driven by occupancy hours	Ventilation (often 100% outside air), exhaust, fume hoods and process equipment, frequently critical
What occupancy control can do	Switch/setback lights and HVAC in confirmed-empty rooms. Korea: 32.4% lighting, 27.9% cooling in 3 classrooms	Reduce non-critical loads (lights, projector, charging, comfort AC). Deep savings require ventilation redesign
Demonstrated savings	27.9–32.4% end-use (S1); 11% education model (S11)	UC Irvine 61% avg / 51% NS-II (S3, S4); ACEEE lab 40% (S11) — achieved by variable air-change-rate control with air-quality and occupancy sensors
Does AIU Smart Campus do this?	Yes — occupancy-confirmed shutdown of non-critical loads under rules	Partly. It shuts down non-critical loads and guards CRITICAL devices; it does not control fume-hood or air-change rates
Planning rate used here	Cons 10% / Mod 18% / High 28% on controlled loads	Cons 5% / Mod 10% / High 20% on a smaller controllable share (40%)

UC Irvine's result is real but comes from reducing laboratory air changes (from roughly one every 5–8 minutes, 24/7, to 2 per hour unoccupied and 4 per hour occupied, per UCI News) using air-quality and occupancy sensing. **We therefore treat 51–61% as evidence that occupancy-aware laboratory ventilation control is powerful — not as a rate our current non-critical-load automation can claim.** If AIU later integrates HVAC/ventilation control for labs, the laboratory rows of the model can be revisited.

5. AIU campus scale

Alamein International University is a national non-profit university in New Alamein City, Matrouh Governorate (inaugurated 2020, presidential decree 435 of 2020). The sources below give three different numbers that are easy to conflate; they refer to different things.

Item	Value	What it actually is	Status	Source
Campus land area	≈150 acres (reported); user-cited '≈150 feddan'	Land area. The sources we accessed say acres; 1 feddan ≈ 4,200 m ² (≈1.04 acre), so 150 acres ≈ 60.7 ha and 150 feddans ≈ 63 ha. Unit to be confirmed with AIU	Not publicly verified (unit)	A1
BUA 263,020 m ²	263,020 m ²	Built-up area of REDCON's contract scope : construction & finishes (Arch & MEP) of 4 educational + 2 housing buildings. Listed 'on-going' on REDCON's site (page © 2023)	Contractor-stated; operational share not verified	A3
BUA 174,000 m ²	174,000 m ²	BUA of SIAC's package ('first stage of the second sector'): Faculty of Arts & Design, Central Library, Student Services, Postgraduate, Administration, Conference Hall, Services + basement	Contractor-stated; one package	A2
BUA 100,390 m ²	100,390 m ²	REC Consultants gives this for the same building list as SIAC . Conflicts with 174,000 m ² — unreconciled	Discrepancy	A4
Phases	3 phases	Redcon phase 1; SIAC phase 2; planned phase 3 = 3 buildings + a major hospital (2018 report)	Phase 3 not quantified	A5

Item	Value	What it actually is	Status	Source
Facilities listed	10 faculty buildings; 4 accommodation towers; central library; administration; university + dentistry hospitals; sports/entertainment area; outdoor theatre; 9 residential buildings outside the campus	Programme description of the campus	Reported (secondary)	A1
Current students	—	—	Not publicly verified	—
Planned capacity	≈15,000 students at full build-out	Master-plan designer statement (30-year plan)	PLANNED	A6

Current / under construction / planned

- **CURRENT (operating):** AIU is open and teaching (faculties launched 2020–2024 per its public profile). The operating built-up area is **not publicly verified**; the contract packages above overlap the operating buildings but cannot be proven to equal them.
- **UNDER CONSTRUCTION:** REDCON lists its project as 'on-going'; the planned third phase (three buildings + hospital) is not quantified.
- **PLANNED / MASTER PLAN:** full build-out for ≈15,000 students; ≈150-acre land. These are never used as built-up area in the model.

Modelling decision

We use **263,020 m²** (the larger single contractor-documented BUA) as the **central reference area**, with **174,000 m²** as a lower sensitivity and **437,020 m²** (263,020 + 174,000, assuming the two packages are additive — unverified) as an upper sensitivity. The first two are never added together in the central case. Replace with AIU's real metered/operational area as soon as it is available.

6. Estimated AIU baseline energy (Method A — energy-use intensity)

Annual energy = built-up area × EUI. Conversion: 1 GWh = 1,000 MWh = 1,000,000 kWh. This is a **benchmark-based estimated baseline, not actual AIU consumption**.

EUI selection (electricity, hot-climate educational buildings)

Case	EUI (kWh/m ² ·yr)	Anchors (from S14 unless noted)	Confidence
Low	130	Roraima Univ. (Brazil, CDD≈3,316) administrative 132; Al-Rajhi 145–155 would be above this	MEDIUM
Mid (central)	165	University of Sharjah ≈165; Dammam 167–217; Al-Rajhi (Saudi) 145–155	MEDIUM
High	215	Dammam upper bound 217; US college median ≈266 kWh/m ² incl. heating fuels (S15) is a cross-check	LOW–MEDIUM

BUA case	Low 130	Mid 165	High 215
174,000 m ² (SIAC package)	22,620,000 kWh = 22.62 GWh	28,710,000 kWh = 28.71 GWh	37,410,000 kWh = 37.41 GWh
263,020 m² (REDCON; central)	34,192,600 kWh = 34.19 GWh	43,398,300 kWh = 43,398 MWh = 43.40 GWh	56,549,300 kWh = 56.55 GWh
437,020 m ² (sum, unverified)	56,812,600 kWh = 56.81 GWh	72,108,300 kWh = 72.11 GWh	93,959,300 kWh = 93.96 GWh

The central baseline of **43.4 GWh/yr** is an order-of-magnitude planning figure. AIU's real consumption may differ materially (hospital loads, central plant, partial occupancy, New Alamein's Mediterranean coastal climate versus Gulf climates).

7. Scenario model

Rather than one number, three scenarios combine three factors. **Annual saved = Baseline × Controllable share × Automation coverage × Saving rate on controlled loads.**

Factor	Conservative	Moderate	High-impact	Evidence / reasoning	Confidence
Controllable share of total electricity	50%	60%	70%	Lighting, HVAC and plug loads that follow occupancy/schedule; excludes lifts, pumps, labs' critical loads, IT, hospital equipment. Cooling ≈70% of electricity in hot-climate campuses (search extract) supports a majority share	LOW
Automation coverage (share of controllable load actually under the system)	50%	70%	90%	A deployment/design target , not evidence. Depends on sensor, actuator and metering rollout	LOW (target)
Saving rate on controlled loads	10%	18%	28%	10%: LBNL median 9% (FDD), ACEEE education 11%. 18%: Pace ≈18%, CMU/DOE target 20%, Crete >20%. 28%: Korea measured 27.9% (cooling) – 32.4% (lighting) and dorm 28.2%	MEDIUM

Why not 30% (or more)?

- 30%+ end-use savings were measured for **specific end-uses over weeks** (S1) or in **whole-building retrofits including HVAC plant/controls changes** (S2: 34%), and 51–61% for lab ventilation redesign. None is a sustained campus-wide result from non-critical-load automation.
- Campus-/portfolio-level evidence clusters at **≈9–20%** (S6, S12, S13). The High-impact case therefore applies 28% only to **controlled** loads and only with 90% coverage.
- **Most defensible range for AIU:** Conservative to Moderate, i.e. **≈5–13% of controllable energy** (≈2.5–7.6% of total). High-impact is a stretch target contingent on real baseline data.

8. AIU savings results

Reference case: $263,020 \text{ m}^2 \times 165 \text{ kWh/m}^2\cdot\text{yr} = 43,398,300 \text{ kWh/yr}$. Five-year values assume flat annual savings from first full year (no ramp-up, degradation, area growth or tariff escalation).

	Conservative	Moderate	High-impact
Controllable energy (kWh/yr)	21,699,150	26,038,980	30,378,810
Under system control (kWh/yr)	10,849,575	18,227,286	27,340,929
Saving rate on controlled loads	10%	18%	28%
Annual energy saved (kWh)	1,084,958	3,280,911	7,655,460
Annual energy saved (MWh / GWh)	1,085 MWh / 1.08 GWh	3,281 MWh / 3.28 GWh	7,655 MWh / 7.66 GWh
Monthly energy saved (kWh)	90,413	273,409	637,955
5-year energy saved (kWh)	5,424,788	16,404,557	38,277,301
% of total campus electricity	2.5%	7.6%	17.6%
% of controllable energy	5.0%	12.6%	25.2%
Projected annual consumption (GWh)	42.31	40.12	35.74

Financial impact (tariff scenario)

Tariff: **EGP 2.55/kWh** — EgyptERA official schedule from April 2026, medium-voltage (22–11 kV) 'other users', excluding the 3 Pt/kWh consumption stamp and service fees. A campus of this size would plausibly be medium-voltage, but **AIU's billing category is not publicly verified**; at low-voltage EGP 2.74/kWh values rise ≈7%.

Constant tariff, no escalation.

	Conservative	Moderate	High-impact
Annual saving (EGP million) @2.55	2.8	8.4	19.5
5-year saving (EGP million) @2.55	13.8	41.8	97.6
Annual saving (EGP million) @2.74 (LV)	3.0	9.0	21.0

CO₂ impact (indicative)

Using Egypt's grid intensity of **0.563 kgCO₂/kWh** (Ember data, lifecycle basis; obtained through a secondary data aggregator — **re-confirm on ember-energy.org before formal citation**; confidence MEDIUM):

	Conservative	Moderate	High-impact
Annual CO ₂ avoided (tCO ₂)	611	1,847	4,310
5-year CO ₂ avoided (tCO ₂)	3,054	9,236	21,550

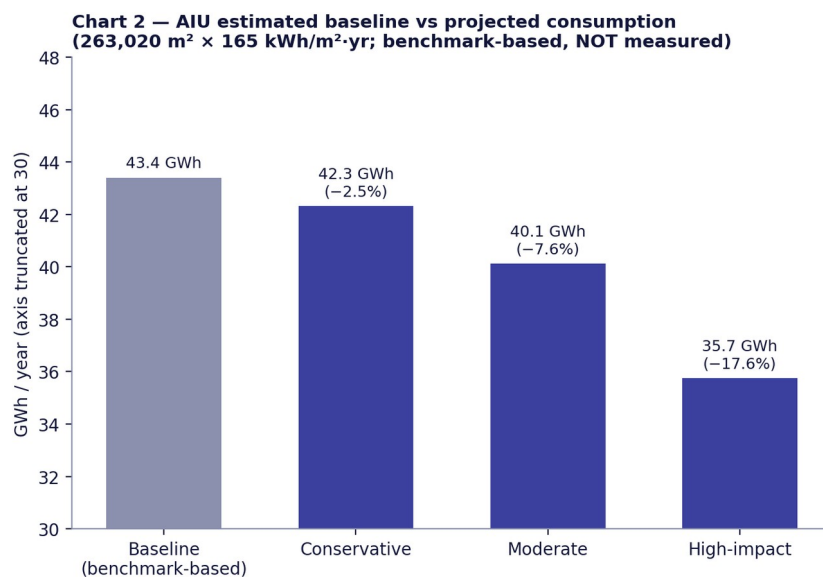


Chart 2 — Baseline vs projected consumption (benchmark-based baseline; axis truncated)

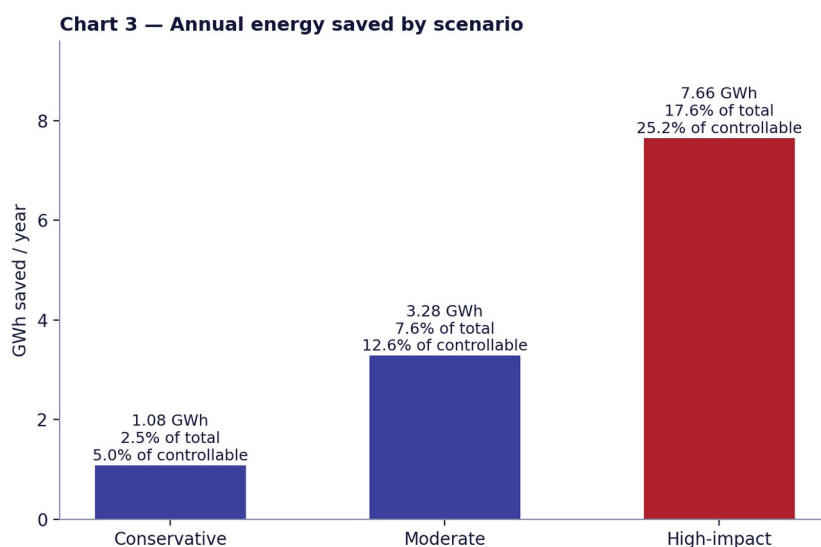


Chart 3 — Annual energy saved by scenario

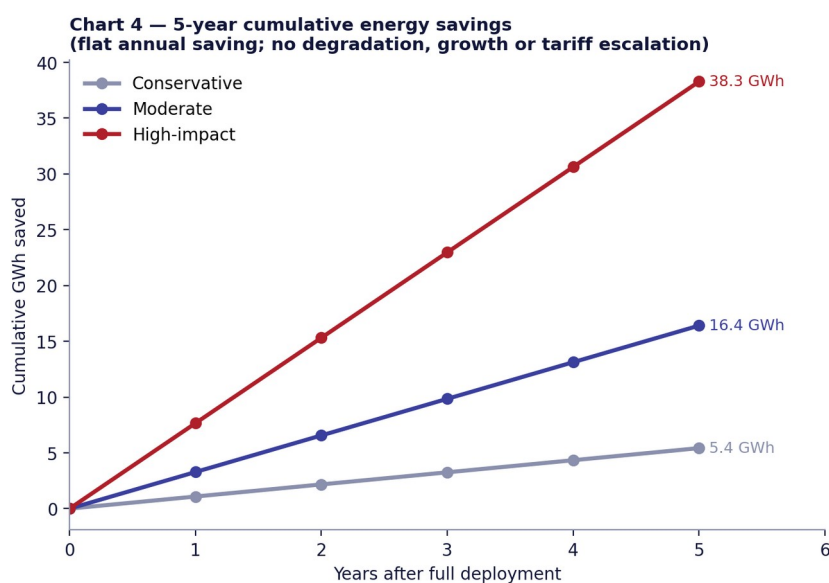


Chart 4 — Five-year cumulative energy saved

Sensitivity: annual savings (MWh) across area and EUI

Area case	EUI 130	EUI 165	EUI 215
174,000 m ²	566 / 1,710 / 3,990	718 / 2,170 / 5,064	935 / 2,828 / 6,599
263,020 m ² (central)	855 / 2,585 / 6,032	1,085 / 3,281 / 7,655	1,414 / 4,275 / 9,975
437,020 m ²	1,420 / 4,295 / 10,022	1,803 / 5,451 / 12,720	2,349 / 7,103 / 16,574

Cells show Conservative / Moderate / High-impact annual savings in MWh. The full range across the sensitivity grid is roughly 0.6 to 16.6 GWh/yr, dominated by area and EUI uncertainty — which is why validation with AIU meter data is the first priority.

9. Building-category estimate

AIU has not published a building-category area breakdown, so shares below are a **clearly labelled scenario assumption (LOW confidence)** applied to the central 263,020 m². Category EUI is held equal (165) to avoid inventing category intensities; categories differ only in controllable share and saving rate (Section 4, Korean results, ACEEE). Hospital and sports facilities are grouped as 'other' with low coverage.

Category	Assumed area share	Baseline (MWh)	Controllable	Control strategy	Saved MWh/yr (C / M / H)
Academic / classrooms	40% (105,208 m ²)	17,359	65%	Occupancy-confirmed lighting/HVAC/plug shutdown	564 / 1,422 / 2,843
Laboratories (teaching)	10% (26,302 m ²)	4,340	40%	Non-critical loads only; critical ventilation guarded	43 / 122 / 312
Admin / library / services	15% (39,453 m ²)	6,510	65%	Schedules + occupancy; plug loads	169 / 444 / 876
Dormitories / housing	25% (65,755 m ²)	10,850	60%	Presence-based room shutdown (cf. Korean RMS)	325 / 820 / 1,640
Sports / hospital / other	10% (26,302 m ²)	4,340	40%	Schedules; limited scope	43 / 122 / 234

Bottom-up cross-check: summing categories gives 1,146 / 2,929 / 5,907 MWh/yr (2.6% / 6.7% / 13.6% of total), against the top-down 1,085 / 3,281 / 7,655 MWh. The two approaches agree within about 6%, 11% and 23% (Conservative, Moderate, High-impact), which supports the planning range but does not validate it. **Academic buildings and dormitories carry the greatest potential** because they combine large area, occupancy-driven loads and the strongest published evidence; laboratories contribute little under non-critical-load control.

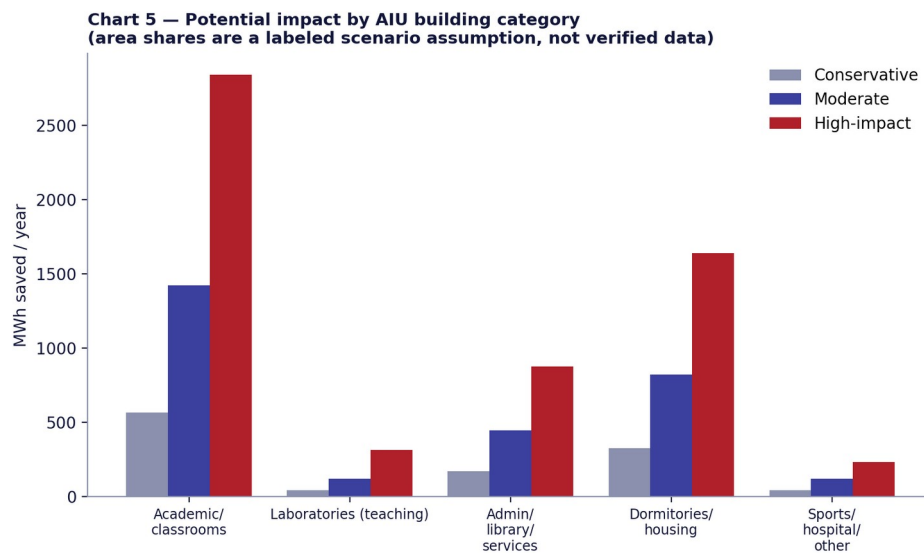


Chart 5 — Potential impact by building category (area shares are assumptions)

10. Mapping the evidence to our system

AIU Smart Campus implements the cycle **SENSE → ANALYZE → VERIFY → DECIDE → ACT → MONITOR** on top of Smart Building, Zones, Sensors, Devices, Automation Rules and Logs, Energy Readings, Access Control, Evidence, Investigation, auditability and a Command Center.

Stage	What our system does	Evidence it relates to
SENSE	PIR occupancy sensing and telemetry feed zone state	Korean sensors (S1); occupancy sensing in S10–S11
ANALYZE	Combines occupancy with the zone schedule (open/closed) and device state	Schedule-aware control; campus platforms (S13)
VERIFY	Confirms a zone is empty for a grace period before acting; avoids shutting down occupied rooms	Addresses false-trigger risk noted in occupancy literature; Korean study had no verification layer
DECIDE	Deterministic rule + confidence; CRITICAL-device protection; fail-safe	Rule-based (not AI/ML)
ACT	Issues commands to non-critical loads; command outcome tracked (success / failed / queued)	End-use control (S1 lighting, EHP)

Stage	What our system does	Evidence it relates to
MONITOR	Energy readings, automation log, honest failure states	Needed for measured, not assumed, savings

Why the architecture can improve on simple occupancy ON/OFF

- **Verification before action** reduces nuisance shutdowns, which is the main reason occupancy control gets disabled by users in practice.
- **Context** (schedule, zone type, critical-load guard) lets the same sensor drive different, safer actions in classrooms, labs and dormitories.
- **Energy readings + automation logs** turn savings from an assumption into a measurable before/after quantity — the validation gap in most published results.
- **Security, evidence and investigation** are not energy features, but they make automated physical actions auditable, which matters for adoption on a real campus.

What the prototype does not yet prove

Our demonstration environment is a software + bench prototype. The current system has **not** measured kWh savings at AIU, has not been deployed campus-wide, and its device commands in the demo environment may report states such as `queued\_no\_broker` or `COMMAND\_FAILED` when no physical broker/hardware responds — we present those honestly. The savings in this report are therefore projections from external evidence.

11. Why the project is valid, and the research gap

This project does **not** claim that smart campuses are a new idea. Existing research shows that occupancy sensing, lighting automation, HVAC automation and room management work, and that university buildings achieve measurable savings (S1–S13).

Our contribution is **integration**: physical sensing + context + verification + automation decision + physical control + energy measurement + security + evidence + investigation + auditability in one cyber-physical campus control platform.

The gap, stated carefully

Published implementations typically concentrate on one or two of lighting, HVAC, occupancy detection, energy monitoring, individual buildings or laboratory optimisation. We did not find, in the sources reviewed, an openly documented university implementation that also ties energy automation to **access control, evidence, investigation and a unified command centre** — but this is a statement about the sources we reviewed, not a claim that no such system exists.

Typical focus in reviewed work	AIU Smart Campus adds
Lighting or HVAC control in one building type	Zone-level automation across rooms/labs with critical-load guard
Energy monitoring dashboards	Closed loop: decision → command → status → energy impact
Occupancy detection accuracy	Verification step and decision logging before action
Energy only	Energy + access control + evidence + investigation + auditability in one Command Center

12. Assumptions and limitations

Value	Source	Why used	Confidence
Reference BUA 263,020 m ²	A3 REDCON (contractor)	Largest single documented BUA; not verified as operational campus total	MEDIUM (as a documented number) / LOW (as campus total)
Sensitivity BUA 174,000 / 437,020 m ²	A2 SIAC; A2+A3 sum	Lower bound and unverified additive upper bound	LOW
EUI 130 / 165 / 215 kWh/m ² -yr	S14 (hot-climate universities)	Electricity-dominant cooling climate analogues; New Alamein climate differs	MEDIUM
Controllable share 50 / 60 / 70%	Engineering estimate	Excludes critical, process and fixed loads	LOW
Automation coverage 50 / 70 / 90%	Project target	Design variable of deployment, not evidence	LOW
Saving on controlled loads 10 / 18 / 28%	S1, S6, S10–S13	Spans campus-level evidence to measured end-use maxima	MEDIUM
Category area shares 40/10/15/25/10%	Engineering estimate	AIU category areas not published	LOW
Lab controllable 40%, rates 5/10/20%	S3–S4, S11 + system scope	Our system does not control lab ventilation	LOW
Tariff EGP 2.55/kWh (2.74 LV)	E1 EgyptERA Apr 2026	Official; AIU category unknown; excludes stamp and fees	MEDIUM
Flat 5-year saving, no escalation	Engineering simplification	Avoids speculative growth, ramp-up or tariff changes	MEDIUM
CO ₂ factor 0.563 kg/kWh	E2 Ember (secondary access)	Egypt-specific grid intensity; lifecycle basis	MEDIUM
Operating hours/semester pattern	—	Not modelled; Korean results come from semester weeks	LOW

Other limitations: published savings are partly weather- and occupancy-dependent; savings from different end-uses are not additive; rebound and behaviour effects are not modelled; capital and maintenance costs are not included (so no payback is claimed); and none of the benchmarks is from Egypt.

13. Final engineering conclusion

- Is there evidence this type of system saves energy in universities?** Yes. A field study with before/after measurements (Seoul) and multiple U.S. programmes show measurable savings from occupancy-based control and automation.
- What range has actually been demonstrated?** 27.9–32.4% on specific classroom end-uses and 28.2% in dormitories (Korea); 9–20% at campus/portfolio level (LBNL, Pace); 34% building-level retrofit (MSU); 51–61% in laboratories through ventilation redesign.
- Defensible target range for AIU?** About **5–13% reduction in controllable energy** (Conservative to Moderate), equal to **2.5–7.6% of total campus electricity**, subject to validation through actual baseline measurements. 25% of controllable energy (High-impact) is an upper bound, not a plan.
- How much energy is that?** On the benchmark-based baseline of 43.40 GWh/yr: **1.08–3.28 GWh/yr** (5,425–16,405 MWh over five years), ≈ EGP 2.8–8.4 million per year and ≈ 611–1,847 tCO₂ per year (indicative). High-impact: 7.66 GWh/yr.
- Which AIU building types have the greatest potential?** Academic/classroom buildings and dormitories, followed by administration/library; laboratories offer less under non-critical-load control; hospital and central plant are largely out of scope.
- What measurements prove the actual AIU savings?** (a) 12 months of utility-meter and sub-meter data to establish the real baseline and the controllable share; (b) AIU's actual tariff category and bills; (c) per-zone energy readings (lighting, AC, plug) with timestamps; (d) occupancy hours by zone; (e) outdoor temperature/degree-days for normalisation; (f) automation-log events linked to measured load drops.

10. **How to validate experimentally?** Replicate the Korean design where it was strongest: alternate-week ON/OFF in matched zones (e.g. Smart Lab 301 and a comparable room), a reference group of untreated rooms with calibration (Eq. 1 of S1), logging at ≥ 15 -minute resolution, ≥ 4 –8 weeks per season, weather normalisation, and reporting results with confidence intervals. Pilot first in one classroom block and one dormitory floor, then extend.

Defensible one-sentence claim

Based on comparable university implementations, AIU could potentially target an estimated **5–13% reduction in controllable energy consumption (≈ 2.5 –7.6% of total campus electricity)**, with a higher-bound scenario of $\approx 25\%$ of controllable energy, subject to validation through actual baseline measurements.

14. Source register

Sources are separated by evidence type. Items that were obtained through secondary extracts, or that we could not independently reopen, are flagged in the last column text.

A. ACADEMIC & OFFICIAL BENCHMARK EVIDENCE

ID	Title	Institution / author	Year	URL	Number extracted	What it represents
S1	Energy Saving of a University Building Using a Motion Detection Sensor and Room Management System	J.-W. Lee & Y. I. Kim, Seoul Nat. Univ. of Science & Technology — Sustainability 12(22):9471	2020	https://doi.org/10.3390/su12229471	77.6% / 32.4% / 27.9% / 28.2%; 918.66 m ² parking, 3 classrooms, 10 dorm rooms vs 120 reference	Measured before/after savings at H University, Seoul (South Korea); parking lighting, classroom lighting, classroom cooling (EHP), dorm (key-card RMS)
S2	Michigan State University: Anthony Hall (Better Buildings Solution Center)	MSU via U.S. DOE Better Buildings (partner-reported; 'not the viewpoint of DOE')	2015	https://betterbuildingssolutioncenter.energy.gov/showcase-projects/michigan-state-university-anthony-hall	34% energy savings; \$536,000/yr; >100,000 sq ft (317,200 sq ft per project summary)	Whole-building (source-EUI) saving after multi-measure controls/HVAC retrofit
S3	UC Irvine: Smart Labs Initiative / Natural Sciences II (Better Buildings Solution Center)	UC Irvine via U.S. DOE Better Buildings	2015	https://betterbuildingssolutioncenter.energy.gov/showcase-projects/uc-irvine-smart-labs-initiativenatural-sciences-ii	51% energy savings; \$180,000/yr	Building-systems energy only (metered VFD + BMS-trend model); excludes lab process loads
S4	Laboratory energy-savings program earns green leadership award (UCI News)	UC Irvine Strategic Communications	2013	https://news.uci.edu/2013/05/29/laboratory-energy-savings-program-earns-green-leadership-award/	61% average total energy saving across 10 retrofitted labs; up to 60%; air changes 2/h unoccupied vs 4/h occupied	Laboratory-specific savings from variable ventilation + air-quality + occupancy sensors
S5	University of California Irvine Smart Labs Initiative Success Story	U.S. DOE Federal Energy Management Program	2018	https://www.energy.gov/cmei/femp/articles/university-california-irvine-smart-labs-initiative-success-story	60% energy savings (headline)	Program headline; DOE video summary
S6	Pace University: Coordinated Energy Management Strategy	Pace University via U.S. DOE Better Buildings	2024	https://betterbuildingssolutioncenter.energy.gov/implementation-models/pace-university-coordinated-energy-management-strategy	26 million kWh + ~1 million therms and \$6.8M saved (through Oct 2022); portfolio 1–5 million sq ft; 24% GHG reduction	Campus-wide program (energy specialists + BMS scheduling + set-point policy); NOT a sensor-hardware result
S7	Pace University partner/poster summary (18% average annual energy reduction)	U.S. DOE Better Buildings (search extract of Summit materials)	2023	https://betterbuildingssolutioncenter.energy.gov/partners/pace-university	~18% average annual energy reduction; 250 million kBtu; \$7.2M	Campus-wide programme average. Secondary extract — re-confirm against the primary poster before citing
S8	Geothermal Heat Pump Case Study: Colorado Mesa University	U.S. DOE (DOE/GO-102025-6426)	2026	https://www.energy.gov/hgeo/geothermal/articles/geothermal-heat-pump-case-study-colorado-mesa-university	14.86 million kWh/yr saved; \$1.5M/yr; 1.2 million sq ft; 16 buildings; 10 kWh/sq ft/yr	Campus district geothermal heat-pump system. A HVAC plant technology, NOT sensing/automation — excluded from the saving-rate range
S9	Better Buildings: University of Utah — Eyring Chemistry & Biology buildings (blog)	U.S. DOE Better Buildings Beat Team	2017	https://betterbuildingssolutioncenter.energy.gov/beat-blog/specialized-savings-better-buildings-university-partner-showcases-big-savings-science	44% EUI saving (estimated); \$456,000/yr; 5 buildings, 370,000 sq ft	Estimated lab/classroom building retrofit (CAV→VAV + sensors/controls)

ID	Title	Institution / author	Year	URL	Number extracted	What it represents
S10	Human-in-the-Loop Sensing/Control for Commercial Building Energy Efficiency and Occupant Comfort	Carnegie Mellon University / U.S. DOE BENEFIT-2016	2017	https://www.energy.gov/cmei/buildings/articles/human-loop-sensingcontrol-commercial-building-energy-efficiency-and	20% energy saving TARGET (occupancy sensing + model-predictive control)	Research target, not a measured result
S11	Smart Buildings: Using Smart Technology to Save Energy in Existing Buildings (Report A1701)	ACEEE	2017	https://www.aceee.org/sites/default/files/publications/researchreports/a1701.pdf	Education (occupancy sensors + web lighting control) 11%; laboratory (air-quality + occupancy sensors + real-time ventilation) 40%; integrated smart building 30–50% where otherwise inefficient	Technology-review model values synthesised from case studies — not a single measured campus
S12	Smart Energy Analytics Campaign — outcomes	Lawrence Berkeley National Laboratory / U.S. DOE	2019–2020	https://www.energy.gov/cmei/buildings/articles/smart-energy-analytics-campaign-reveals-continued-energy-and-cost-savings	Median 9% (FDD) and 3% (EIS); 104 organisations, 6,500 buildings, >0.5 billion sq ft	Portfolio-wide median savings from analytics software (office + higher-education = 80% of participants)
S13	Energy management of campus buildings (CAMP-IT), cited as ref. [12] in S1	D. Kolokotsa et al., Technical University of Crete (Greece) — Energy and Buildings	2009 (via S1)	https://doi.org/10.3390/su12229471 (S1 literature review)	15–30% depending on period of year (as reported in S1); >20% efficiency per secondary search	Web-based campus energy-management platform. Taken from S1's literature review — original paper not independently opened
S14	Benchmarking of Energy Consumption in Higher Education Buildings in Saudi Arabia: Sulaiman Al-Rajhi University Case (Table 2 compiles worldwide university EUIs)	Energies 16(3):1204	2023	https://www.mdpi.com/1996-1073/16/3/1204	Al-Rajhi 145–155 kWh/m ² ·yr; Sharjah ≈165; Dammam 167–217; Roraima admin 132 kWh/m ² ·yr	Measured/literature electricity EUIs of university buildings in hot climates
S15	U.S. National Median EUI by Property Type (ENERGY STAR Portfolio Manager)	U.S. EPA / ENERGY STAR (CBECS-based)	2024	https://portfoliomanager.energystar.gov/pdf/reference/US%20National%20Median%20Table.pdf	College/University site EUI 84.3 kBtu/ft ² (≈266 kWh/m ² , all fuels)	US median incl. heating fuels — used as an upper cross-check only; not climate-matched to Egypt

B. OFFICIAL AIU / CONTRACT DATA

ID	Title	Institution / author	Year	URL	Number extracted	What it represents
A1	Alamein International University (encyclopaedic summary of AIU's own site)	Wikipedia citing aiu.edu.eg (archive 2023)	2025	https://en.wikipedia.org/wiki/Alamein_International_University	~150 acres; central library; outdoor theatre; sports area; university hospital; dentistry hospital; administration building; 10 faculty buildings; 4 accommodation towers; 9 residential buildings outside campus	Describes the built/operating campus programme (area stated in ACRES in this source). Secondary — confirm on aiu.edu.eg
A2	Alamein International University of Science and Technology — first stage of the second sector	SIAC Construction (contractor)	2026 (page)	https://www.siac.com.eg/?q=experience-sectors%2Fbuilding%2Falamein-international-university-science-and-technology-first-stage	Total built-up area 174,000 m ² ; Faculty of Arts & Design, Central Library, Student Services, Postgraduate, Administration, Conference Hall, Services Building + basement	BUA of ONE contract package (phase/sector), not the whole campus
A3	Alamein International University (project page)	REDCON Construction (contractor)	2023 (page)	https://redconcon.com/project/alamein-university/	BUA 263,020 m ² ; client NUCA; scope: construction & finishes (Arch & MEP) for 4 educational + 2 housing buildings; marked 'on-going'	BUA of REDCON's contract scope; not necessarily the total campus

ID	Title	Institution / author	Year	URL	Number extracted	What it represents
A4	Alamein International University of Science and Technology (project page)	REC Consultants (structural consultant)	2026 (page)	https://rec-consultants.net/alamein-project	Total BUA 100,390 m ² with the SAME building list as A2 (A2 says 174,000 m ²)	Unreconciled discrepancy between two firms for the same package — shown as sensitivity, not resolved
A5	Egypt to launch third work phase for New Alamein University in 2019	Amwal Al Ghad	2018	https://en.amwalalghad.com/egypt-to-launch-third-work-phase-for-new-alamein-university-in-2019/	3 phases; Redcon = phase 1; SIAC = phase 2 (six administrative buildings + utilities); phase 3 = 3 buildings + a major hospital; New Alamein City 48,000 feddans	News report (2018) used to place contract packages in phases; planned 3rd phase is NOT quantified
A6	Alamein University Master Plan	Architizer (designer listing)	n/d	https://architizer.com/projects/alamein-university-master-plan/	Master plan for 30 years; full build-out for ≈15,000 students	PLANNED / master-plan student capacity (designer statement, not an enrolment figure)

C. ENGINEERING ESTIMATES (inputs from external sources)

ID	Title	Institution / author	Year	URL	Number extracted	What it represents
E1	EgyptERA — Electricity Tariff Starting From April 2026	Egyptian Electric Utility & Consumer Protection Regulatory Agency	2026	https://egyptera.org/en/TariffApril2026.aspx	Medium voltage (22–11 kV) 'other users' 255 Pt/kWh; low voltage 274 Pt/kWh; commercial >1000 kWh 279 Pt/kWh; consumption stamp 3 Pt/kWh (non-residential)	Official tariff schedule (excl. service fees/stamp unless stated). AIU's actual billing category is not publicly verified
E2	Egypt grid electricity intensity 0.563 kgCO ₂ /kWh	Ember (via GreenCalculus data table 'Ember 2025')	2025	https://greencalculus.com/data/grid-electricity-emission-factors/	0.563 kgCO ₂ /kWh (lifecycle basis)	Used for an indicative CO ₂ estimate; accessed through a secondary aggregator — re-confirm on ember-energy.org

D. OUR PROJECT TARGET

Item	Value	Basis
Automation coverage	50 / 70 / 90% of controllable load	Deployment target for the platform (not evidence)
Planning range	5–13% of controllable energy (≈2.5–7.6% of total)	Conservative-to-Moderate scenarios derived above; to be validated
Stretch scenario	≈25% of controllable energy (≈17.6% of total)	High-impact: Korean-level rates at 90% coverage

Appendix A. Benchmark data sheet

Fields requested for every benchmark. 'n/s' = not stated in the source we accessed; nothing has been filled in by assumption.

Benchmark	Building / campus	Area	# bldgs	Energy consumption	Occup. sensing	HVAC ctrl	Light ctrl	Energy monitoring	Automation	Saving %	Annual kWh saved	Annual cost saved	Method	Pub. date
H Univ. Seoul (S1)	Cyber Bldg; dorm; B1 parking	16,476 m ² ; 5,994.53 m ² ; 918.66 m ² test area	2 + parking	Sensor-off: 50.9 (parking), 77.1 light / 197.1 cooling (class) Wh/m ² -day; dorm ref. 25.3 kWh/wk	Yes (RF/Doppler; key-card RMS)	Yes (EHP)	Yes	Yes (metered experiments)	Sensor-triggered	77.6 / 32.4 / 27.9 / 28.2	n/s (39.5; 25.0; 55.0; 142.4 Wh/m ² -day)	n/s	Measured ON/OFF alternate weeks; dorm reference-group calibration	14 Nov 2020

Benchmark	Building / campus	Area	# bldgs	Energy consumption	Occup. sensing	HVAC ctrl	Light ctrl	Energy monitoring	Automation	Saving %	Annual kWh saved	Annual cost saved	Method	Pub. date
MSU Anthony Hall (S2)	Anthony Hall, E. Lansing	317,200 sq ft	1	n/s (EUI blank on page)	Partial (demand ventilation, lighting controls)	Yes	Yes	Airflow monitoring	Controls	34% (source EUI)	n/s	\$536,000	Partner-reported	22 Oct 2015
UC Irvine NS-II (S3)	Natural Sciences II	> 100,000 sq ft	1	n/s	Yes	Yes (ventilation)	Yes	Yes (VFD meters + BMS trends)	Yes	51%	n/s	\$180,000	Metered VFD data + BMS-trend model; excludes lab process loads	22 Oct 2015
UC Irvine program (S4,S5)	10 retrofitted labs	n/s	10 labs	n/s	Yes	Yes (air changes)	n/s	Yes	Yes	61% avg; up to 60% (S4); 60% (S5)	n/s	n/s	University-reported average	29 May 2013; 4 Oct 2018
Pace University (S6,S7)	NYC / Westchester portfolio	1–5 million sq ft	n/s	n/s	Tag only	Yes (match HVAC to occupancy)	n/s	Yes (BMS sweeps, energy accounting)	Scheduling; no capex	≈18% avg (secondary)	26,000,000 kWh + ~1M therms (through Oct 2022)	\$6.8M	Weather/size-normalised expected vs actual	26 Mar 2024
Colorado Mesa (S8)	Grand Junction campus	1.2 million sq ft	16	10 kWh/sq ft-yr	No	Geothermal plant	No	n/s	No	≈ half of conventional HVAC	14,860,000	\$1.5M	DOE case study	May 2026
Univ. of Utah Eyring (S9)	Chemistry & Biology	370,000 sq ft	5	n/s	Sensors (type n/s)	Yes (VAV)	n/s	Yes	Yes	44% EUI (estimated)	n/s	\$456,000 (est.)	Estimate	5 Apr 2017
CMU / DOE BENEFIT (S10)	Research buildings	n/s	n/s	n/s	Yes (depth sensing)	Yes (MPC)	n/s	Yes	Yes	20% TARGET	n/s	n/s	Research target	18 Oct 2017
LBNL SEA (S12)	104 organisations	> 0.5 bn sq ft	6,500	n/s	No	FDD	n/s	Yes (EIS/FDD)	Analytics	9% FDD / 3% EIS median	n/s	n/s	Year-2 vs baseline medians	2019–2020
ACEEE (S11)	Model education / lab bldgs	100,000 / 70,000 sq ft	model	190,000 / 980,000 kWh	Yes	Lab: real-time ventilation	Yes (education)	Yes	Yes	11% / 40%	20,900 / 390,000	n/s	Review of case studies	2017